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Large-Scale Assessment of Binding Free Energy Calculations in Active Drug Discovery Projects

Christina E. M. Schindler,* Hannah Baumann, Andreas Blum, Dietrich Böse, Hans-Peter Buchstaller, Lars Burgdorf, Daniel Cappel, Eugene Chekler, Paul Czodrowski, Dieter Dorsch, Merveille K. I. Eguida, Bruce Follows, Thomas Fuchß, Ulrich Grädler, Jakub Gunera, Theresa Johnson, Catherine Jorand Lebrun, Srinivasa Karra, Markus Klein, Tim Knehans, Lisa Koetzner, Mireille Krier, Matthias Leiendecker, Birgitta Leuthner, Liwei Li, Igor Mochalkin, Djordje Musil, Constantin Neagu, Friedrich Rippmann, Kai Schiemann, Robert Schulz, Thomas Steinbrecher, Eva-Maria Tanzer, Andrea Unzue Lopez, Ariele Viacava Follis, Ansgar Wegener, and Daniel Kuhn*

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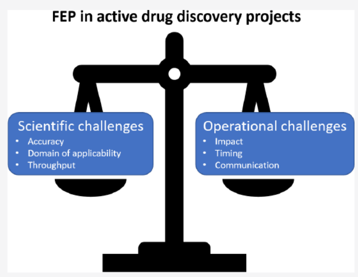
ABSTRACT: Accurate ranking of compounds with regards to their binding affinity to a protein using computational methods is of great interest to pharmaceutical research. Physics-based free energy calculations are regarded as the most rigorous way to estimate binding affinity. In recent years, many retrospective studies carried out both in academia and industry have demonstrated its potential. Here, we present the results of large-scale *prospective* application of the FEP+ method in active drug discovery projects in an industry setting at Merck KGaA, Darmstadt, Germany. We compare these prospective data to results obtained on a new diverse, public benchmark of eight pharmaceutically relevant targets. Our results offer insights into the challenges faced when using free energy calculations in real-life drug discovery projects and identify limitations that could be tackled by future method development. The new public data set we provide to the community can support further method development and comparative benchmarking of free energy calculations.



■ INTRODUCTION

Identifying ligands that bind with high affinity to a target protein, while balancing other ligand properties relevant to safety and biological efficacy, is the goal of small-molecule drug discovery projects. To support this challenging enterprise, accurate prediction of protein−ligand binding free energies has long been a goal of computer-aided drug design (CADD). Physics-based free energy calculations using Monte Carlo or Molecular Dynamics simulations are considered the most rigorous approach to this problem. Yet, until recently, widespread usage was hindered by high computational costs, limitations of sampling algorithms, and limitations in (small molecule) force fields. In addition, from an industry perspective, using free energy calculations was considered challenging due to the limited automation, throughput, and robustness of the protocols.

Over the last 10 years, however, there has been tremendous progress in sampling,[1−8] force field development,[9−18] throughput,[19−28] and automation.[5,14,29−41] As a result, many pharmaceutical companies have adopted relative free energy calculations[42] and, in particular, the Schrödinger FEP+ workflow[5] as a new computational tool to support their drug discovery efforts. In 2016, we started a large initiative at Merck KGaA, Darmstadt, Germany to thoroughly assess the prediction accuracy and identify the best use cases of free energy calculations. In this initiative, we aimed to apply FEP+ in all suitable in-house active drug discovery projects prospectively. There were three main reasons for this particular setup of testing a new method in a real-life prospective application. First, blind assessment of computational tools—as in the form of community wide prediction challenges such as CASP,[43] CAPRI,[44] SAMPL,[45] and D3R Grand Challenges[46]—has been demonstrated to provide a more realistic picture of the accuracy and limitations of a given method. Therefore, in this initiative we focused on applying FEP+ prospectively, i.e., in a blind manner. Second, it was unclear how much time constraints, limitations on resources, and information gaps that are prevalent in real-life drug discovery projects affect the performance of the method relative to what was reported previously in the literature.[5,17,47,48] Third, since a large initial assessment of the FEP+ method in 2015,[5] many features had

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been added to the method, e.g., calculating free energies for transformations involving ring openings and net charge changes.[47,48] These challenging types of transformations are often used in drug discovery projects but were not present in the previous benchmark data set.

Here, we present retrospective and prospective data collected in 18 in-house drug discovery projects over three years. We describe the general workflow we established for using free energy calculations in projects and report the performance of the FEP+ method on in-house targets. We further present data collected from a new diverse benchmark and discuss key learnings for domain of applicability, project impact, and limitations of the method.

■ RESULTS AND DISCUSSION

Free energy calculations—in particular Schrödinger's FEP+ implementation—have emerged as an accurate binding affinity prediction method that could potentially accelerate hit-to-candidate optimization. While the theoretical accuracy of the method is determined by the accuracy of the force field used and the extent of sampling, in practice, system setup and structure preparation also affect the performance. It is therefore not clear whether the previously reported accuracy of this method[5] can be maintained when it is applied under the constraints of an industry setting and how it can best contribute to compound design and the related multiparameter optimization process. Therefore, in 2016, a large initiative was launched at Merck KGaA, Darmstadt, Germany to thoroughly evaluate and prospectively benchmark the FEP+ technology in active projects. From 2016 to 2019, we applied FEP+ prospectively to 12 targets and 23 chemical series performing over 35,000 individual perturbation calculations. We finally obtained valid predictions for over 6,000 chemical entities. More than 400 blindly predicted and novel molecules were synthesized and tested. This yielded a large set of prospective data providing a detailed assessment of the method's accuracy in a typical small molecule drug discovery working environment. In addition to benchmarking, this initiative also enabled us to define best practices and explore optimal use cases.

Free Energy Calculation Workflow in Projects. Throughout the last three years, we established a workflow for deploying free energy calculations in projects. The process is shown schematically in Figure 1. First, we assess the general feasibility of using FEP for a given target and chemical series of interest by collecting available structural data and experimentally determined binding affinities. At this stage, we typically require at least one well-resolved cocrystal structure with one representative of the series of interest. This strict requirement is based on our experiences in three projects where we tried to use homology models in the absence of an X-ray structure. In contrast to previous successful applications of FEP with homology models,[49−51] we obtained unsatisfactory prediction accuracy retrospectively in all three cases (RMSE > 1.5 kcal/mol). In two of the three projects, we later had access to a cocrystal structure and were then able to successfully complete the validation phase (see "FEP Feasibility and Validation Results in in-House Drug Discovery Projects"). While we had no success in using homology models for FEP, we had mixed experience using binding mode hypothesis obtained by docking in the absence of a cocrystal structure for the series of interest. We encountered this scenario in two projects during our FEP evaluation. In both cases, we attempted the validation study based on a docking

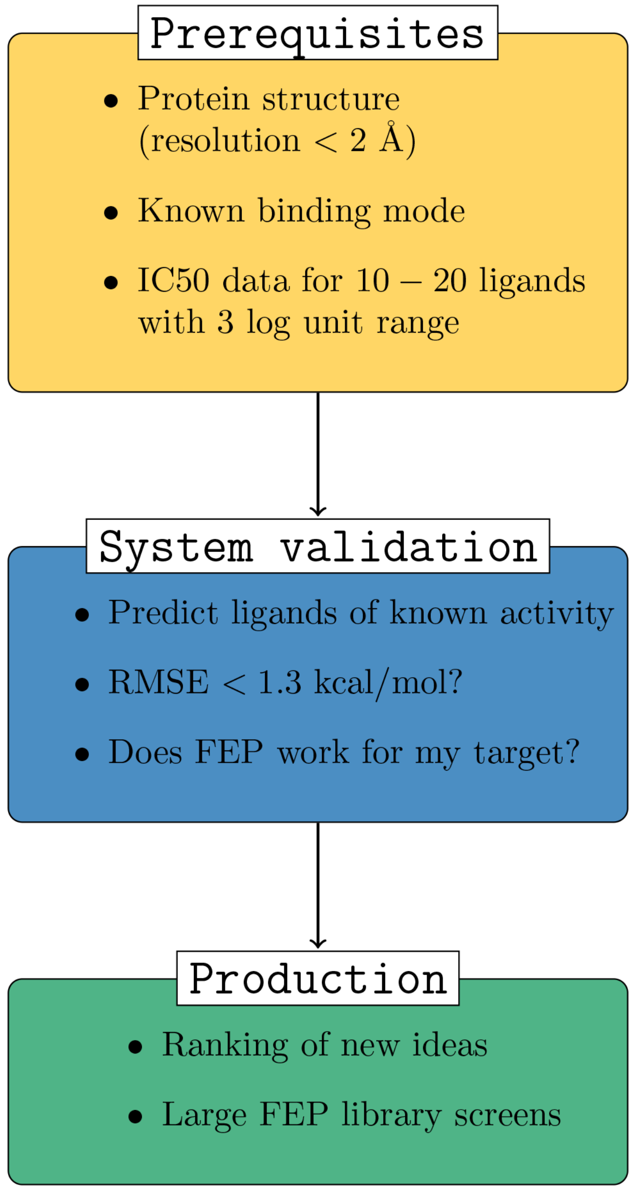

Figure 1. Deploying free energy calculations in drug discovery projects at Merck KGaA, Darmstadt, Germany.

pose obtained using a cocrystal structure of another series in the first case and the apo structure of the protein in the second. In the first project, we were not able to obtain good agreement with experimental data. The cocrystal structure that was obtained later showed considerable protein flexibility in the binding site, although the predicted binding pose was relatively similar to the crystal structure. In a second project, we were indeed able to obtain good agreement between predicted and experimental affinities using a docking pose as a model system (Figure S1(A)). Later, we performed a second validation study using the cocrystal structure that was solved in the mean time. This confirmed the good prediction accuracy on a larger and structurally more diverse set of ligands (Figure S1(B)). The docking pose was very similar to the binding mode shown by the cocrystal structure.

Once we have sufficient structural data available, we collect a data set of congeneric ligands with experimental binding affinities (at least 10 ligands, preferably 20) and all available information on the biochemical and biophysical assays (e.g.,

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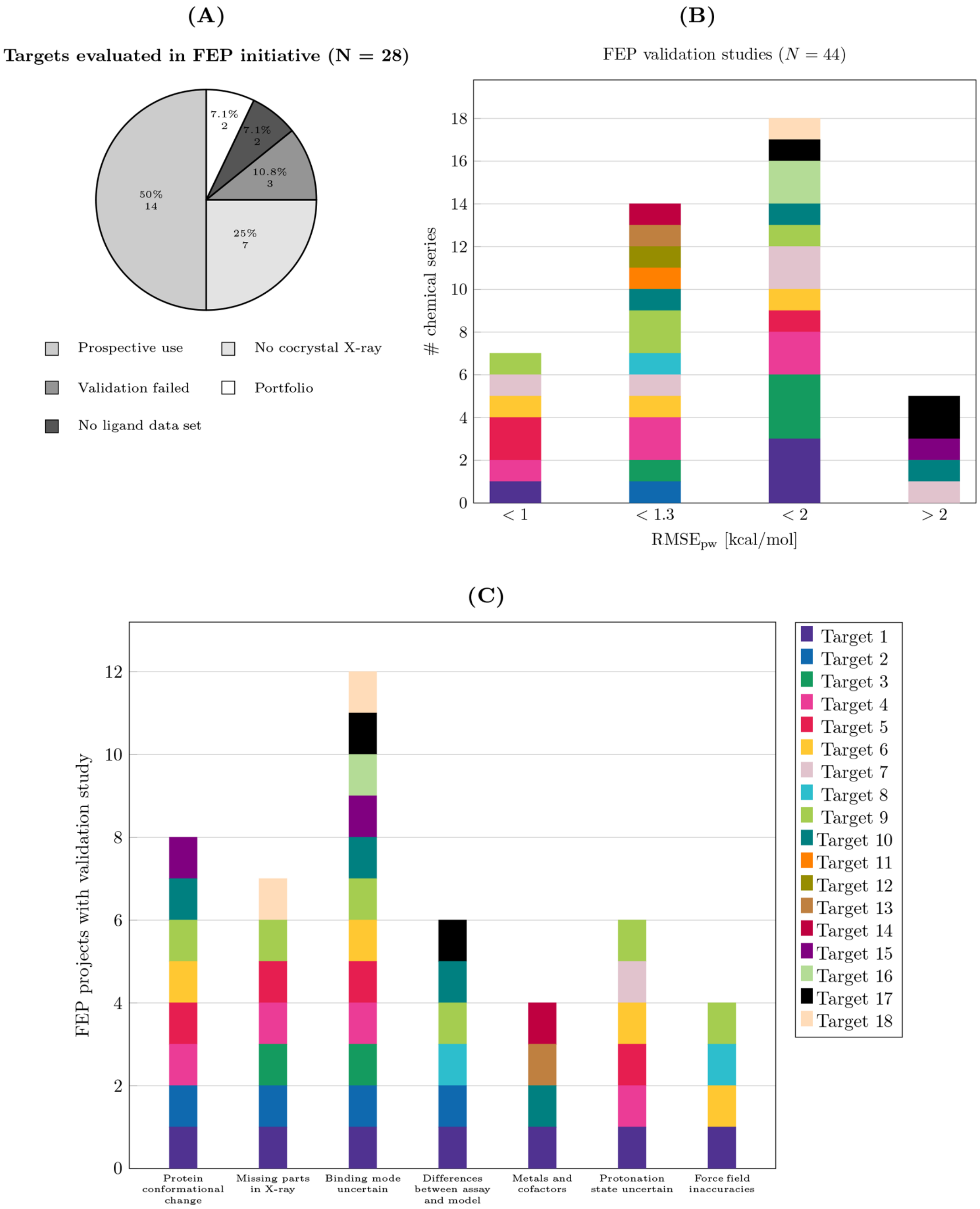

Figure 2. FEP feasibility, validation results, and challenges in in-house projects. (A) Outcome of FEP evaluation on 28 targets. The main reason for not being able to use FEP in projects was the lack of structural data. (B) Results for validation studies using FEP+. In total, validation studies were performed for 44 chemical series. Different colors represent different target receptors (the legend is shown below). For many targets, multiple chemical series were evaluated. Accuracy varies greatly depending not only on the target but also on the chemical series of interest. (C) Possible challenges for prediction accuracy are encountered in all projects. The projects are colored by target receptors as in (B).

protein construct, buffer conditions, presence of cofactors, etc.). Note that the recommendation for the size of the ligand data set is rather a "rule-of-thumb", typically it is very difficult to obtain larger data sets in early stage projects. If the ligand data set is sufficiently large, it can be split according to the R-groups of the molecule that are modified to identify potentially different accuracy for predictions in different parts of the binding site. Then based on these data sets, retrospective free energy calculations are performed in order to compare predicted to experimentally determined binding affinities. We typically refer to these retrospective computational experiments as validation studies. The accuracy of the free energy calculations is assessed by calculating the pairwise root-mean-square error ($RMSE_{pw}$) between relative predicted affinities $\Delta\Delta G_{pred}$ and relative experimental affinities $\Delta\Delta G_{exp}$ for all ligand pairs in the set. $\Delta\Delta G_{pred}$ is calculated by subtracting the predicted values ΔG_{pred} (including cycle closure correction). If available, predicted affinities are compared to experimental results from different assays. Typically, during the validation phase, different input structures and settings are evaluated in order to find the best setup for later prospective calculations. In practice, time constraints often limit this phase to evaluating typically 3 possible model systems. Model systems may vary in terms of the input structures, the sampling time, and the

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sampling settings (e.g., number of λ-windows). Variations in the input structures may involve the crystal structure used if multiple structures of the protein and/or the series of interest have been solved, the binding mode (different docking poses or different poses compatible with the electron density), the part of the protein structure used (single domain or multiple domains if applicable), the oligomeric state of the protein if an oligomer observed in the crystal structure has a known biological function, or whether possible protein−protein interaction partners of the target protein are considered.

In the case of large outliers ($|\Delta G_{pred} - \Delta G_{exp}| > 2$ kcal/mol), detailed analysis is done in order to understand the underlying causes. We usually consider a validation study as successful if we achieve an $RMSE_{pw}$ smaller than 1.3 kcal/mol, and if we are able to sufficiently explain large outliers if present. While the exact accuracy needed for FEP depends on the application and accuracy up to 2 kcal/mol is typically considered useful for scoring large libraries,[52] we have found that this stricter threshold of $RMSE_{pw} < 1.3$ kcal/mol in the validation study gave us a better chance of achieving sufficient accuracy (RMSE < 2 kcal/mol) in the prospective setting. During production mode, prediction accuracy typically deteriorated compared to the validation study (see below).

If the dynamic range of the data set is suitable, FEP predictions should also yield good ranking. In reality, however, we frequently encountered the situation that only a data set with a limited dynamic range was available at the time of the validation study, which made model evaluation challenging.[53] In such cases, we proceeded to the prospective phase in a "trial" fashion. We predicted all molecules that were being synthesized without using the predictions for prioritization and evaluated these prospective predictions to assess the accuracy of the method. Based on these results, we decided whether or not to apply FEP to the project in production mode.

After successfully completing validation stage, the FEP project moves into production mode, and prospective calculations are performed for compound ideas. These ideas have to be sufficiently similar to the validated chemical series. In the case of a new chemical series and new available structural information, a new validation study has to be performed. We closely monitor the prospective accuracy of the predictions throughout the project and track which predicted compounds have been synthesized. All predicted affinities and structures are stored in a database. Using an automated workflow, we periodically check our in-house database to detect whether predicted compounds have been synthesized and tested in the meantime. In our experience, this constant monitoring of synthesized molecules and hence prospective prediction accuracy is necessary to initially build trust in the project team and later detect when the chemical matter has moved outside of the domain of applicability of the model. Structure matching is done first based on exact structure matches. If no match is found, matching is performed based on matching tautomers or matching without considering stereochemistry to maximize the number of data collected. A tag specifying the "matching mode" is assigned to the final result, and all data that have been matched based on tautomers or without considering stereochemistry are manually checked. In the case of multiple isomers, one prediction is matched to multiple entries and activities in the database, and the correct one has to be assigned by visual inspection. We assume that matching single isomer predictions to the respective racemate does not result in a large error compared to the typical error of free energy methods. However, it is clear that this might not be valid for all projects, and such assumptions should be taken into account when analyzing prediction accuracy. Still, for our data set, the largest prospective outliers did not involve any molecules with defined stereochemistry that were only synthesized as racemate (see "Prospective FEP+ Results in in-House Projects").

FEP Feasibility and Validation Results in in-House Drug Discovery Projects. Over the course of three years, we evaluated 28 targets for general FEP feasibility (Figure 2(A)). We performed validation studies for 18 targets and 44 chemical series and progressed 14 targets and 25 chemical series to prospective calculations (as mentioned before, initially our criteria for considering a validation as successful were more lenient). Note that we define chemical series as they were used in the projects; this is, however, project-specific and often depends on the chemists involved. The major reason for not being able to perform a validation study for a given target was the lack of relevant structural data (7 targets). For three targets, all validation studies were considered unsuccessful, and for another two targets, the projects were stopped shortly after a FEP feasibility evaluation or validation study was done (portfolio category). Some chemical series were not progressed to production mode despite good validation results, as they had been deprioritized in the meantime by the project team for other reasons. Overall, we observed a relatively low attrition rate for potential FEP targets, once enough structural and binding affinity data became available to perform a validation study.

The accuracies achieved in the validation study for 18 targets are shown in Figure 2(B). In total, we were able to obtain high accuracy ($RMSE_{pw} < 1$ kcal/mol) and acceptable accuracy ($RMSE_{pw} < 1.3$ kcal/mol) predictions for 14 targets and 21 chemical series. At earlier stages in the initiative, we also accepted validation studies with an $RMSE_{pw}$ larger than 1.3 kcal/mol as successful and therefore progressed these series to production mode. During the course of our evaluation, we tightened the criteria for a successful validation study, since lower accuracy ($RMSE_{pw}$ between 1.3 and 1.6 kcal/mol) in validation studies consistently led to even lower accuracy in a prospective setting. Prediction accuracy varied not only between different targets but also between different chemical series for the same target protein. Most notably, for one target, we obtained results for multiple chemical series covering the whole range from high accuracy ($RMSE_{pw} < 1$ kcal/mol) to low accuracy predictions ($RMSE_{pw} > 2$ kcal/mol) (Figure 2(B), light pink color). Furthermore, when using FEP in active projects, we regularly faced challenges that might affect the accuracy of the method. A qualitative assessment of these challenges is shown in Figure 2(C). Almost all projects had at least one aspect that might be problematic for applying free energy calculations (Figure 2(C)): unsurprisingly, real-life drug discovery projects are not ideal case scenarios. The most common challenges we encountered in the projects where we attempted a validation study were uncertainties in the binding mode of at least a subset of the ligands and uncertainties in the protein structure due to suspected protein conformational change (66% and 44%, respectively). In six projects, we found that the source of experimental data had an influence on whether we could consider the validation study successful. In one case, we initially compared the predicted affinities to the output of a functional assay and found large deviations. However, when comparing the same predicted affinities to

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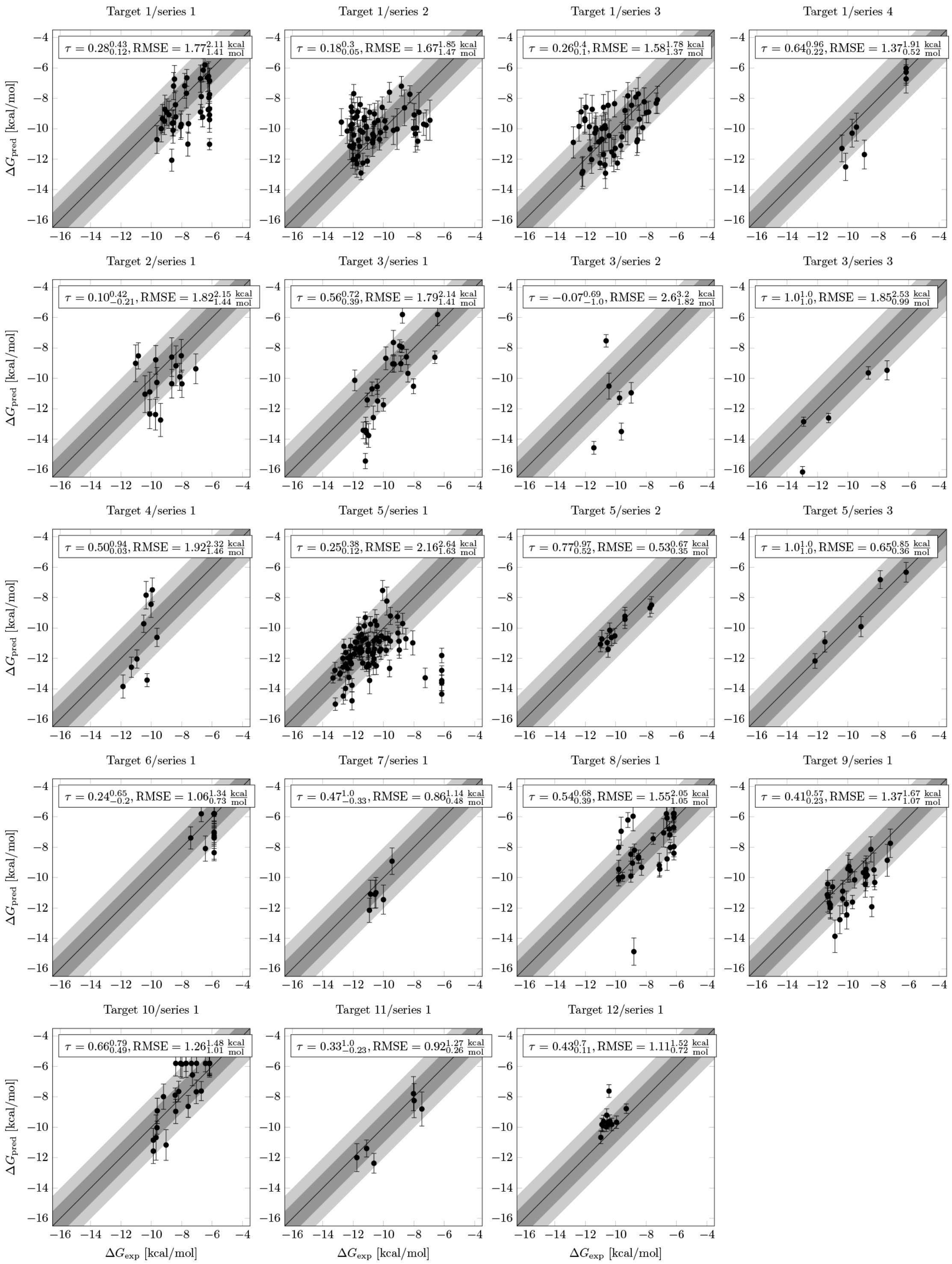

Figure 3. Prospective FEP+ results from 12 targets and 19 chemical series. Experimental affinities were converted to ΔG values using the equation $\Delta G_{\text{exp}} \approx k_B T \log \text{IC}_{50}$. FEP+ accuracy varies between different targets and chemical series. Predictions within 1 and 2 kcal/mol of the experimental affinity are highlighted by dark and light gray areas, respectively. Further performance measures including confidence intervals for each data set are listed in Table S1.

biophysical surface plasmon resonance (SPR) data, we found good agreement and therefore decided to progress the series to production mode (Figure S2). On the one hand, it is expected that FEP results should, in general, correspond best to biophysical binding data and could display larger errors when compared to other types of assays. On the other hand, to have a strong impact in projects, FEP predicted affinities should correlate well to the main assay that is used to drive compound optimization. Also, in many projects, there is only a limited amount of biophysical data available compared to data from biochemical or functional assays. We would discourage investing resources to perform a validation study in projects where the main optimization assay does not correlate well with

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biophysical data, since the impact of FEP on the project will likely remain limited.

In four projects, we found that the parameters of the small molecule force field might not have described the interactions accurately. In two of these projects, using a later version of the OPLS3e force field improved the accuracy for the ligand set used in the validation study (data not shown). In one project, the change in the force field was related to the partial charges. In the second project, the compound had a substituted aliphatic ring. Reparametrization of torsion potentials in this ring as done in recent versions of the Force Field Builder helped to improve the accuracy of the predictions. For the three projects where we found large outliers for the prospective predictions; however, recalculation with newer versions of the force field did not change the results (these cases are discussed in the following section).

Note that here we listed *potential* aspects that may affect the accuracy of the predictions. However, we were not generally able to pinpoint the probability of success for a validation study to a specific feature. For example, we analyzed whether the resolution of the X-ray structure had an influence on the accuracy achieved in the validation study but found no correlation (data not shown).

Prospective FEP+ Results for in-House Projects. For 12 targets and 19 chemical series, it was possible to obtain a prospective data set consisting of at least five data points. The results for each target and series are shown in Figure 3 (see also Table S1 for further performance measures and confidence intervals calculated by bootstrap sampling for each data set). We report performance measures as $x_{x_{low}}^{x_{high}}$ where x is the mean statistic calculated from the complete data set (e.g., RMSE), and x_{low} and x_{high} indicate 90% confidence intervals (see the Experimental Section).

Compared to the results from the validation study, we observed a lower accuracy in prospective applications as indicated by higher RMSE values. Note that here we measured accuracy by comparing predicted free energy ΔG_{pred} to experimental affinity ΔG_{exp} for each compound individually instead of calculating errors for relative free energies $\Delta\Delta G$ for all ligand pairs as in the validation study. The reason for this was that the ligands did not originate from a single FEP map but from multiple independent runs, and in some cases, different molecules were used as reference structures. There were two main reasons for using different reference molecules throughout the project. First, we wanted to stay as close as possible to the novel chemical space that we screened with FEP to avoid large transformations and potentially allow for better performance of the method. The second reason was operational: we wanted to include the molecules that were at that time of highest interest to the chemist. Note that using different experimental reference points might be problematic for accuracy evaluation of a relative free energy method. This is especially important for cases where different reference molecules cover largely different affinity ranges. For example, for two maps that were calculated using a reference molecule with high affinity and low affinity, respectively, the relative ranking of the maps will already be perfect based on the reference molecules (if predicted free energies do not show large changes relative to the reference molecules). In such a case, an evaluation of the full data set will lead to an overestimation of the prediction accuracy. For our cases, however, even dissimilar reference molecules that were used were in a similar affinity range.

For three (albeit small) data sets, we obtained high accuracy predictions (RMSE < 1 kcal/mol), for four sets, we obtained medium accuracy (RMSE < 1.5 kcal/mol; this is the accuracy we expect to see for a series that showed RMSE < 1.3 kcal/mol in the validation study), and for another eight sets, we obtained acceptable accuracy predictions (RMSE < 2 kcal/mol). Binding affinity prediction methods with moderate accuracy of RMSE < 2 kcal/mol are considered useful for scoring larger libraries.[52] We were able to achieve this moderate accuracy level in 17 out of 19 chemical series from our prospective applications. For 13 out of the 19 sets, we either found good correlation (Kendall $\tau > 0.5$) or low error (RMSE < 1.5 kcal/mol). In Figure 3, the targets are ordered chronologically. This figure therefore also indicates an "FEP learning curve": the RMSE for the first five targets is higher than the RMSE calculated for the most recent five data sets (RMSE = $1.78_{1.4}^{2.12}$ kcal/mol for 324 ligands and RMSE = $1.35_{0.97}^{1.72}$ kcal/mol for 118 ligands, respectively). Due to the large confidence intervals, it is, however, difficult to make a clear assessment of improvements.

We investigated the reasons for the largest outliers in the data set. For the analysis, we reset the predicted affinities of all compounds that were predicted to be above the top or below the bottom of the assay to these values and then evaluated the resulting deviation from the experimental value. Based on this analysis, we still identified 23 compounds where the predicted affinity deviated by more than 3 kcal/mol from the experimental affinity. For four of these 23 outliers that originated from target 1/series 1, the validation study had previously shown larger errors for modifications in this part of the molecule. Based on the stricter criteria established later in the FEP initiative, we would not have progressed this project to prospective phase. Another four compounds from target 1/series 2 and target 1/series 3 displayed changes in aromatic heterocycles relative to the reference compound. Initially, we used OPLS3 force field and the deviation from experiment was even larger (absolute error >5 kcal/mol). When rerunning these with the OPLS3e force field, this was reduced to ≈3 kcal/mol. We also recalculated the compounds from target 1/series 2 with Schrödinger suite version 2020-1; however, this did not change the results (see Figure S3 for the resulting minimal example map). While the relative affinity between two of the compounds was captured within the typical error range of the method, the affinity of these two compounds relative to the third scaffold was clearly underestimated.

For three compounds from target 2/series 1 and target 5/series 1, we suspect that the changes relative to the molecule that was cocrystallized could have invoked protein conformational changes that were not captured in the simulations (the changes were made in a part of the binding site that displayed flexibility in the available crystal structures). For a group of five outliers from target 5/series 1, an additional domain that was not part of the crystallization construct might have caused the sudden drop in potency when making modifications to the solvent-exposed side of the molecule (the five molecules were found to be inactive in the assay despite having relatively small modifications compared to highly active molecules). For the crystal structure, a construct containing only the domain with the active site was used (covering roughly one-third of the full-length protein). In contrast, the binding affinity assay used the full-length protein. All of these five outlier molecules were overpredicted with FEP by more than 5 kcal/mol, contributing significantly to the overall RMSE of $2.16_{1.63}^{2.64}$ kcal/mol observed for this data set. When excluding these five molecules, the RMSE calculated over the remaining 83 compounds was reduced to $1.38_{1.12}^{1.66}$ kcal/mol.

Finally, two of the outliers that originated from target 8 and target 9 had modifications to a sulfone or a sulfonamide. When we initially performed the FEP+ calculations for these molecules, we noticed that many modifications were predicted to be favorable when replacing the sulfone or the sulfonamide group (which already seemed "suspicious" at the time). Indeed, when we synthesized the best predicted compounds in these two projects, we found that the affinity was overpredicted by more than 4 kcal/mol. Repeating this calculation for target 8 with a minimal example map in Schrödinger version 2020-1 still showed this large overestimation (data not shown). We also identified a similar case in our public benchmark for the HIF2-α data set (the transformation is shown in Figure S4(A)). These examples might hint at a problem of the OPLS3/OPLS3e small molecule force field in describing the properties of sulfones in relatively hydrophobic binding pockets. Recently, Jorgensen and coworkers proposed improvements to the fixed charged model in OPLS force fields to better capture the properties of sulfur-containing molecules.[54] However, it might be necessary to include polarizability to fully capture the protein–ligand interactions of such sulfur-containing groups.[55]

In addition, there are several general reasons that could account for the larger errors in the prospective data sets compared to the validation studies. First, for several targets, the prospective data set is larger than the original validation set. The larger sample size results in a more robust estimation of RMSE. Second, throughout the course of a project, new ideas tend to be less similar to the chemical space that was used in the validation study and less similar to the representative of the chemical series that was captured in the crystal structure. This results in higher uncertainties with regards to the binding mode and protonation state of the ligands. These uncertainties might, in turn, affect the accuracy of the calculations if incorrect input poses are chosen and sampling is limited. Careful preparation of the input structures and good 3D alignment are important for obtaining accurate free energy estimates with FEP+, since the automatic map generation relies on the input structures. Initially, we used Flexible Ligand Alignment to place the novel molecules into the binding site. However, neglecting the protein yielded sometimes suboptimal structures. In our hands, using Glide core-constrained with MCS or custom SMARTS gave better starting poses with lower need to manually adapt the poses (which is also important in an industry context). This is in agreement with a recent publication that systematically investigated the impact of different alignment methods on FEP+ predictions.[56] In many cases, after the initial alignment, there were still uncertainties about the binding mode, e.g., for asymmetrically substituted rings. In such cases and for uncertainties in tautomers or charge states, we either simulated all possible states in the map when ranking the full set of compounds. Alternatively, we first tried to predict the relevant state with FEP+ (binding pose FEP) and then ranked the whole set based on the predicted optimal state for each compound. Recently, there has been a possibility to include multiple protonation states in FEP+.[57] However, cases in which the protein changes its protonation state upon binding or cases where the ligand has a different pK_a in the binding site than in the solvent[58−60] still cannot be treated properly. Constant pH simulations have made great progress over the past decade,[61−66] yet at the moment, these methods appear too computationally demanding for an industry context.

Third, we found that the diverse chemical matter in our in-house compound collection remained a challenge for small molecule force fields. For almost every new chemical series, we had to reparametrize some of the torsion potentials, even when using the new OPLS3e force field that includes a very high number of torsion potentials.[17] New concepts like the SMIRNOFF "Parsley" force field developed by the Open Force Field Consortium[67] appear promising in helping to more easily extend the force field for novel chemistry and determine suitable parameters. We anticipate that small molecule force fields will continue to improve and better capture the properties of different chemical groups in the future. This, in turn, will hopefully reduce errors in free energy calculations.

Fourth, when using free energy calculations prospectively, we tend to focus on extreme predictions (e.g., the top-ranked compounds). This bias in focusing on extreme predictions can be softened by, e.g., applying a selection bias correction.[68] However, in our hands this did not affect the largest outliers much.

Finally, many transformations that were of high interest to the project teams were innately challenging for the method (e.g., from aromatic ring systems to aliphatic chains, charge changes, addition of new groups via flexible linkers). Especially, the addition of a new functional group with a flexible linker is difficult in terms of sampling, but it frequently occurs in the context of early hit optimization and in fragment optimization.

When comparing ranking by FEP+ to ranking obtained from Glide docking, Glide rescoring, and Prime molecular mechanics generalized Born surface area (MMGB-SA) scoring (see the Experimental Section for details), we found that overall FEP+ performed better than these standard structure-based drug design (SBDD) methods (Table S2). Cohen's d for Kendall τ was 0.85 for comparison to Glide docking, 0.67 for Glide rescoring, and 0.58 for Prime, respectively, indicating a medium effect size.[69] Note that for comparison with Glide rescoring and Prime, Target 6/series 1 was not considered for calculating effect size and also that for technical reasons it was not possible to rank all ligands with the two other methods (e.g., some ligands failed to dock into the protein or minimize due to clashes or gave positives scores in Prime MM-GBSA and were therefore excluded from the analysis).

In four cases (target 1/series 4, target 4/series 1, target 5/series 3, and target 6/series 1), Prime MM-GBSA appeared to yield good rankings (Kendall $\tau > 0.5$) with similar or higher correlations than that of FEP+ based ranking. In two of these cases (target 1/series 4 and target 4/series 1), this is also true for $\text{Glide}_{\text{score}}$. However, since the data sets are all very small (<10 ligands) and therefore confidence intervals are very large, it is difficult to draw a final conclusion on the relative performance of the different methods. Still, this might hint at the opportunity to use simpler scoring methods for certain cases and focus with a computationally expensive method like FEP+ on those cases where other methods fail to rank the ligands.

We also compared FEP+ performance on our prospective in-house data sets to ranking with simple descriptors like molecular weight and log P (correlation statistics are shown in Table S3). FEP+ also outperformed these "null models" as

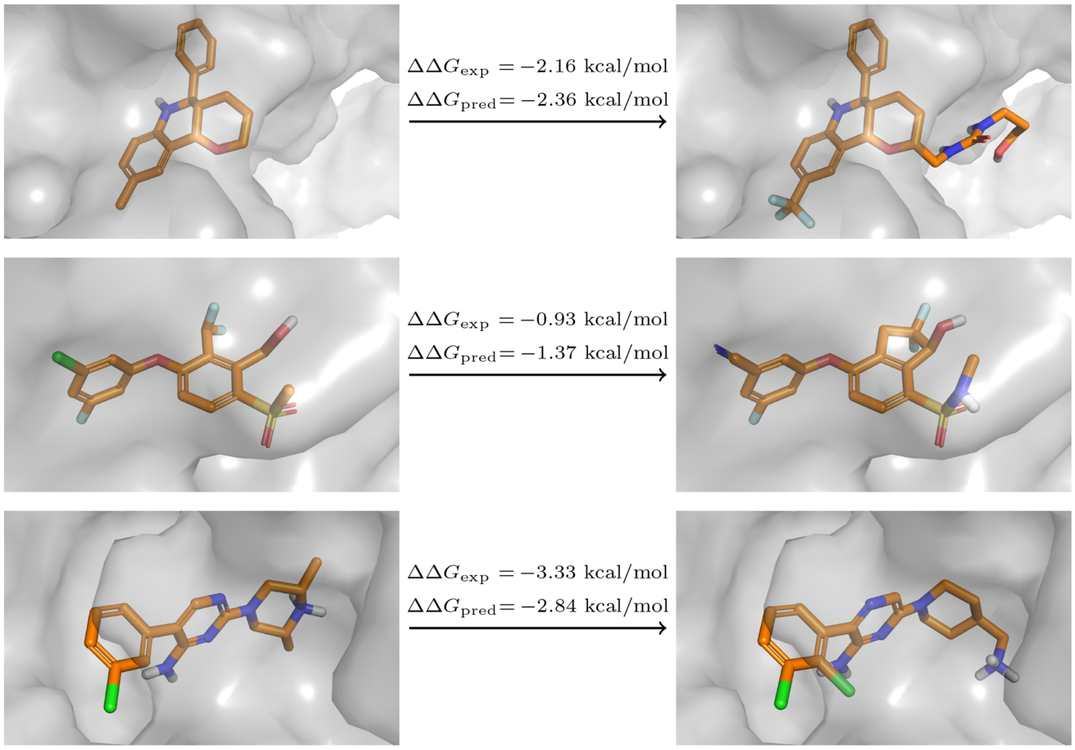

Figure 4. Examples of different types of transformations in the new benchmark set. (A) Addition of a flexible chain in Eg5 leads to increased binding affinity. (B) Ring closure transformation in HIF-2α. FEP correctly predicts an increase in potency, likely due to the reduced ligand flexibility. (C) Shift of a charged amine in SHP-2.

Table 1. Results for Ranking with FEP+ on a Literature Curated Benchmark Set[a]

			Maximum			FEP+			
protein	N	PDB ID	R^2	ρ	τ	R^2	ρ	τ	$\mathrm{RMSE}_{\mathrm{pw}}$
CDK8	33	5HNB	$0.96_{0.94}^{0.97}$	$0.97_{0.95}^{0.98}$	$0.87_{0.83}^{0.91}$	$0.38_{0.18}^{0.67}$	$0.74_{0.56}^{0.86}$	$0.57_{0.41}^{0.72}$	$2.09_{1.98}^{2.19}$
c-Met	24	4R1Y	$0.97_{0.95}^{0.98}$	$0.98_{0.96}^{0.99}$	$0.90_{0.86}^{0.94}$	$0.81_{0.7}^{0.89}$	$0.88_{0.74}^{0.94}$	$0.73_{0.59}^{0.84}$	$1.43_{1.34}^{1.51}$
Eg5	28	3L9H	$0.91_{0.86}^{0.94}$	$0.94_{0.9}^{0.97}$	$0.81_{0.74}^{0.87}$	$0.5_{0.33}^{0.69}$	$0.72_{0.52}^{0.84}$	$0.54_{0.39}^{0.68}$	$1.23_{1.15}^{1.31}$
HIF-2α	42	5TBM	$0.93_{0.9}^{0.95}$	$0.94_{0.9}^{0.96}$	$0.8_{0.75}^{0.85}$	$0.37_{0.1}^{0.65}$	$0.59_{0.36}^{0.77}$	$0.45_{0.27}^{0.61}$	$1.6_{1.52}^{1.69}$
PFKFB3	40	6HVI	$0.96_{0.94}^{0.97}$	$0.98_{0.96}^{0.99}$	$0.89_{0.85}^{0.92}$	$0.63_{0.5}^{0.75}$	$0.79_{0.67}^{0.86}$	$0.6_{0.49}^{0.69}$	$1.78_{1.72}^{1.84}$
SHP-2	26	5EHR	$0.92_{0.88}^{0.96}$	$0.94_{0.9}^{0.97}$	$0.81_{0.75}^{0.88}$	$0.5_{0.32}^{0.69}$	$0.78_{0.59}^{0.88}$	$0.61_{0.45}^{0.74}$	$1.39_{1.3}^{1.47}$
SYK	44	4PV0	$0.92_{0.89}^{0.94}$	$0.92_{0.87}^{0.95}$	$0.78_{0.72}^{0.82}$	$0.25_{0.04}^{0.49}$	$0.42_{0.17}^{0.62}$	$0.29_{0.11}^{0.46}$	$1.61_{1.54}^{1.67}$
TNKS2	27	4UI5	$0.95_{0.92}^{0.97}$	$0.96_{0.94}^{0.98}$	$0.85_{0.8}^{0.9}$	$0.16_{0.01}^{0.41}$	$0.41_{0.07}^{0.66}$	$0.29_{0.05}^{0.51}$	$2.2_{2.08}^{2.32}$
total	264		$0.94_{0.91}^{0.96}$	$0.95_{0.92}^{0.97}$	$0.84_{0.78}^{0.88}$	$0.44_{0.25}^{0.64}$	$0.65_{0.44}^{0.79}$	$0.50_{0.33}^{0.64}$	$1.68_{1.59}^{1.76}$

[a]Calculations were run for 5 ns per λ-window (default sampling time). Pairwise root-mean-squared-error $\mathrm{RMSE}_{\mathrm{pw}}$ was calculated for predicted and experimental $\Delta\Delta G$ values of all ligand pairs. $\Delta\Delta G$ values were derived from the predicted free energy ΔG/score. Averages were calculated weighted by the number of the ligands in the respective data set. For comparison, the maximally achievable performance ("Maximum") based on the known experimental error of the assay[53] is listed (see the Experimental Section for details). Comparison of FEP+ results with simpler ranking methods can be found in Supplemental Tables S5 and S6.

indicated by a large and medium effect size (Cohen's d for Kendall $\tau = 1.03$ and 0.64 for comparing FEP+ with ranking by molecular weight and log P).[69] Glide docking showed small effect size compared to ranking by molecular weight and performed no better than ranking by log P (Cohen's d for Kendall $\tau = 0.36$ and 0.06, respectively). Glide rescoring showed small effect size (Cohen's d for Kendall $\tau = 0.47$ and 0.21, respectively), and Prime MM-GBSA scoring showed medium and small effect size (Cohen's d for Kendall $\tau = 0.57$ and 0.2, respectively).[70]

Benchmark Results. Based on our extensive experience with free energy calculations in our in-house projects where we unfortunately cannot publish the structures at this moment, we decided to construct a new benchmark consisting of eight challenging, recently published data sets with pharmaceutically relevant targets.[71−80] The benchmark comprises 264 ligands in total. The protein targets and the chemical space in this benchmark data set are representative of the targets in our in-house projects. There is, however, no overlap in the ligand series with the in-house data sets. The type and size of alchemical transformations reflect the type and size of chemical changes during hit-to-lead and lead optimization. Overall, this data set illustrates many of the challenges we faced in projects well. In contrast to a previously published benchmark,[5] the transformations includes changes in the net charge and the charge distribution of the molecules as well as ring openings and core hopping (examples are shown in Figure 4). The ligand sets also display a slightly increased range of structural

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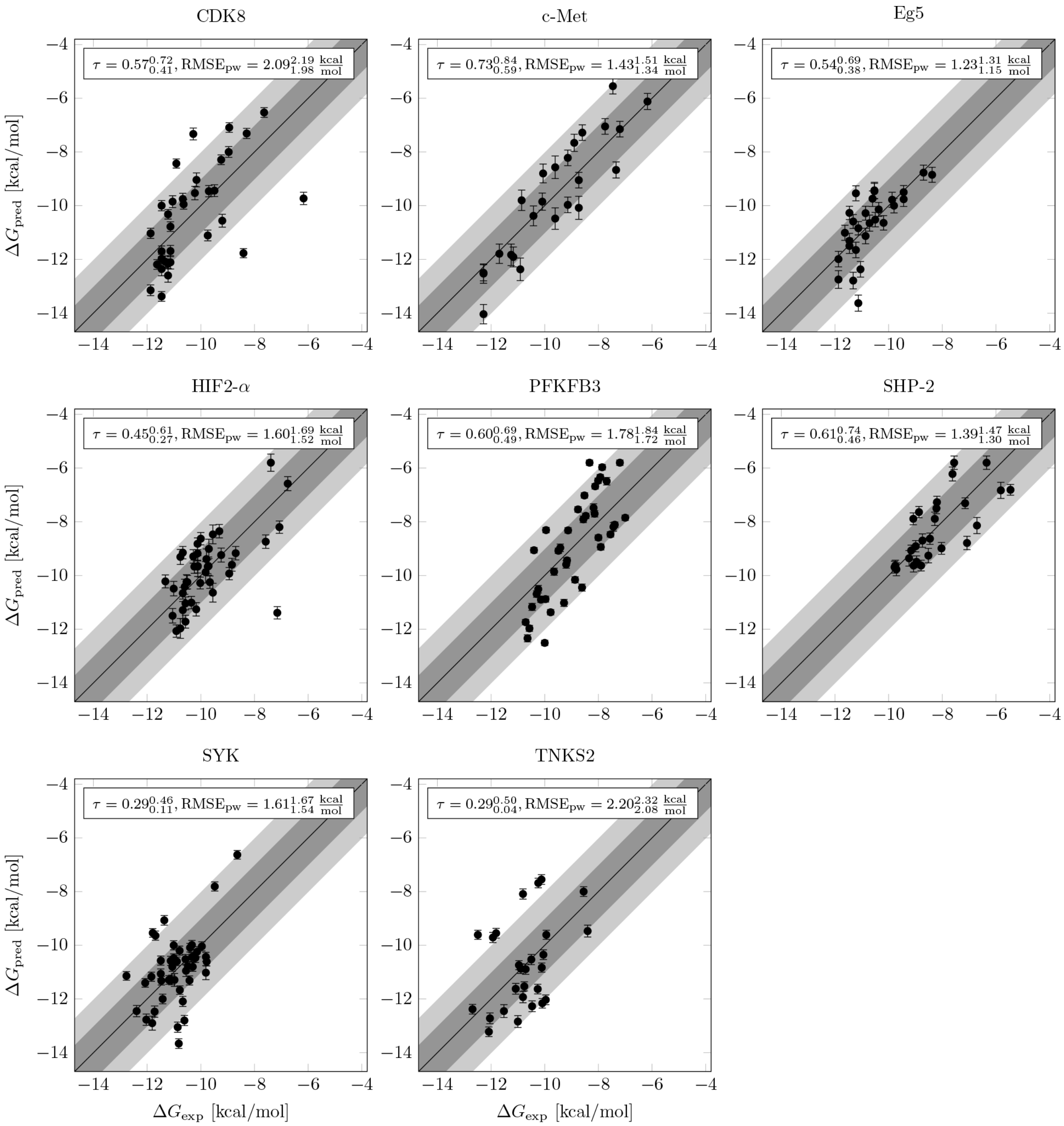

Figure 5. FEP+ results on a literature curated benchmark. For each data set, experimental and predicted ΔG values are shown. Calculations were run for 5 ns per λ-window. Experimental affinities were converted to ΔG values using the equation $\Delta G_{\mathrm{exp}} \approx k_{\mathrm{B}}T \log \mathrm{IC}_{50}$. Predictions within 1 and 2 kcal/mol of the experimental affinity are highlighted by dark and light gray areas, respectively.

diversity compared to the previous benchmark (average maximal mean pairwise Tanimoto similarity $0.79^{0.87}_{0.68}$ vs $0.84^{0.92}_{0.76}$ using the RDKit Daylight fingerprint with default settings). The FEP+ results are summarized in Table 1 (detailed plots for each set can be found in Figure 5). Overall, we achieve good correlation for these data sets (average $R^2 = 0.44^{0.64}_{0.25}$, average Kendall $\tau = 0.50^{0.64}_{0.33}$). The root-mean-square error for the relative affinities $\Delta\Delta G$ calculated over all ligand pairs in the map $\mathrm{RMSE_{pw}}$ ranges between 1.2 and 2.1 kcal/mol. Note that here we use $\Delta\Delta G$ to evaluate RMSE in contrast to the prospective data set where we used ΔG. The reason for this is that FEP+ uses all experimental free energies ΔG_{exp} to calculate the offset used for deriving the predicted affinities ΔG_{pred},[5] and we wanted to avoid an overestimation of the performance. Given the challenges that are included in this data set, this is a remarkable achievement. However, compared to the accuracies reported earlier on a large benchmark set[5] and for internal drug discovery projects at Schrödinger (RMSE = 1.1 kcal/mol, Lingle Wang, personal communication), we see considerably worse performance. This lower accuracy is in line with the prospective accuracy found in our in-house projects (average $\mathrm{RMSE_{pw}} = 1.68^{1.76}_{1.59}$ kcal/mol and average RMSE = $1.64^{1.97}_{1.26}$ kcal/mol, respectively). When analyzing the error for the different types of transformations (Figure S5), we found that transformations involving changes of the net charge or the charge location or changes to the core/scaffold of the molecule showed lower accuracy. These transformations are already considered as inherently more difficult by the FEP+

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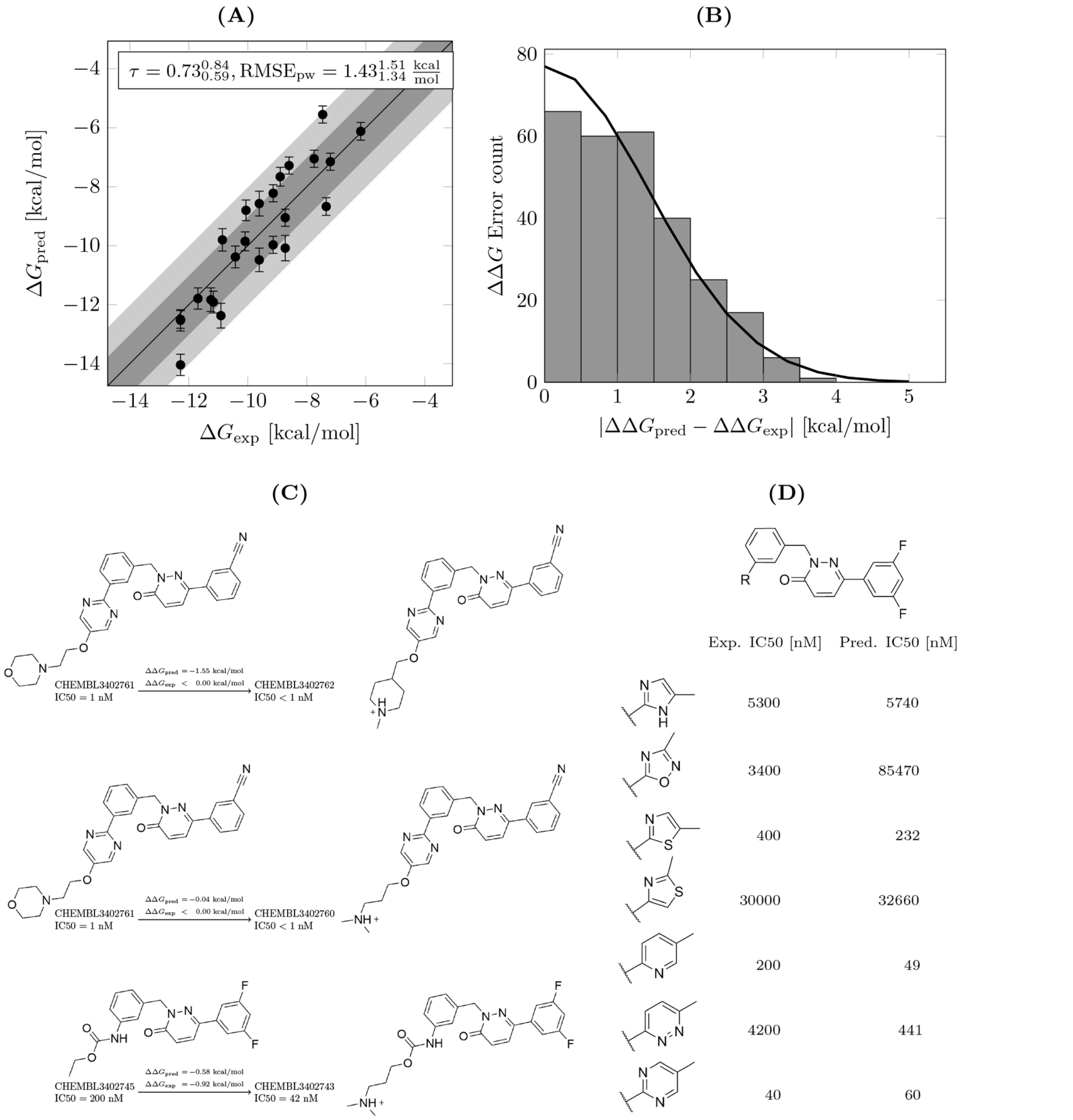

Figure 6. FEP+ results for the c-Met benchmark case. (A) Predicted affinities ΔG_{pred} correlate well with experimental affinities ΔG_{exp}. (B) Histogram of the errors for pairwise relative affinities $|\Delta\Delta G_{\mathrm{pred}} - \Delta\Delta G_{\mathrm{exp}}|$. The solid line shows a Gaussian curve with a standard deviation of 1.43 kcal/mol. (C) Perturbations involving net charge changes in the data set. FEP+ shows good accuracy in predicting these challenging transformations. (D) FEP correctly predicts ranks of a series of different aromatic heterocycles substitutions.

software, and special settings for sampling are used.[47,48] Still, these transformations display larger errors. Interestingly, for the remaining transformations, we could not observe a strong dependency of the error on the size of the transformation, i.e., the number of heavy atoms that are changed. This has also been recognized in an earlier evaluation by Christ and Fox.[36] While it might seem counterintuitive at first, this likely reflects the fact that the number of perturbed heavy atoms is not a precise measure of the "difficulty" of a given transformation. Defining a better measure is, however, an endeavor beyond the scope of this study.

It is interesting and quite sobering to note that based on our guidelines for a successful validation study (good correlation and $\mathrm{RMSE_{pw}} < 1.3$ kcal/mol), we would have only been able to progress one out of the eight chemical series in the benchmark to production mode based on the data obtained for the default setting of 5 ns sampling time. Increasing the simulation time from 5 to 20 ns per λ-window decreased the $\mathrm{RMSE_{pw}}$ for seven out of eight targets by more than 0.05 kcal/mol and decreased the average $\mathrm{RMSE_{pw}}$ by 0.13 kcal/mol (average $\mathrm{RMSE_{pw}} = 1.51^{1.73}_{1.44}$ kcal/mol, see Table S4). This increase in quantitative agreement between predicted and experimental affinities had, however, no effect on ranking/correlation (average Kendall $\tau = 0.49^{0.65}_{0.32}$). Nevertheless, based on the 20 ns data, we would have been able to progress three out of the eight targets to production mode.

To investigate whether more recent versions of FEP+ show improved performance on the benchmark, we recalculated PFKFB3 and SYK data sets using Schrödinger version 2020-1. We obtained slightly improved performance for PFKFB3 ($\tau =$

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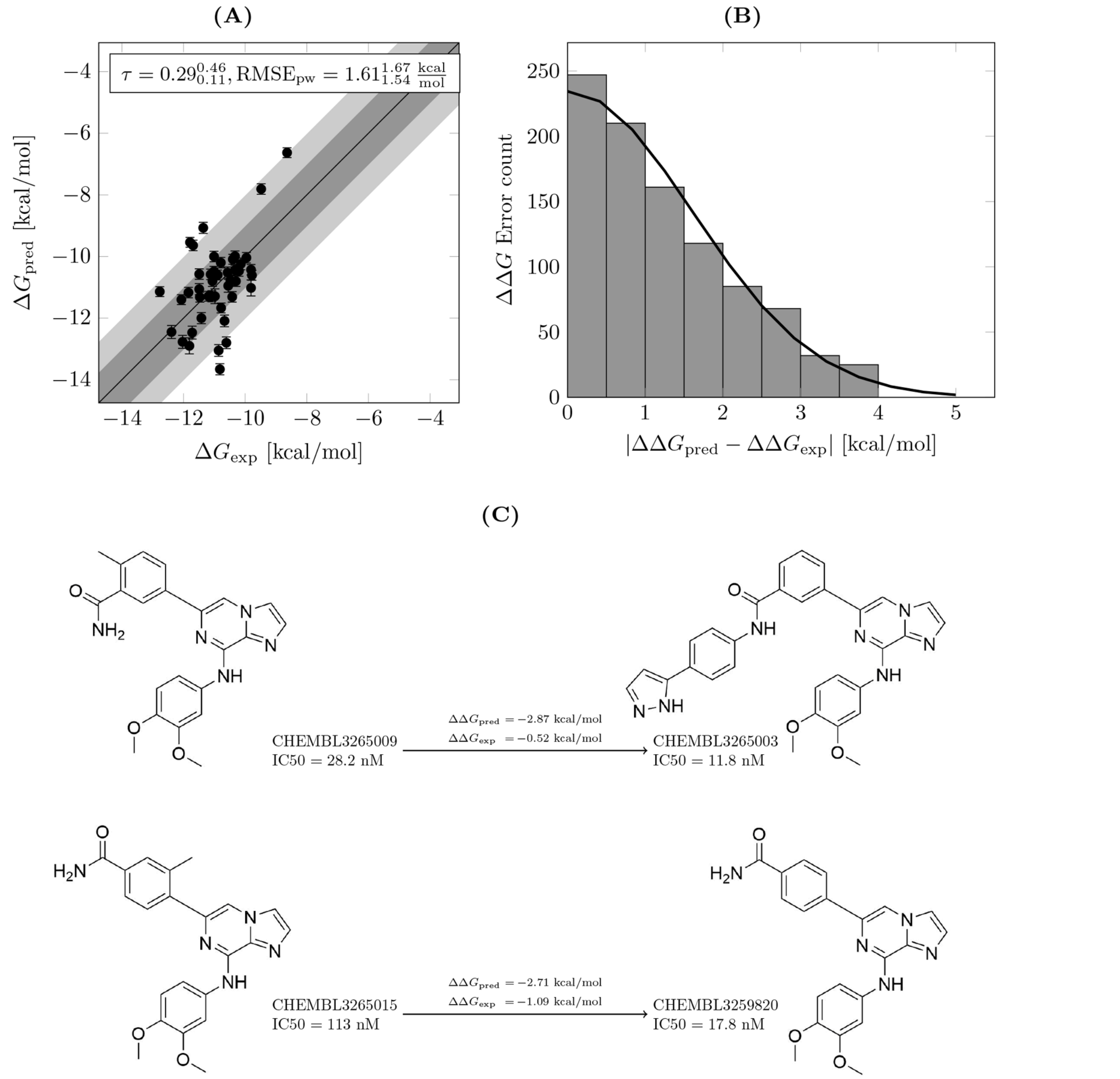

Figure 7. FEP+ results for the SYK benchmark case. (A) Predicted affinities ΔG_{pred} do not correlate with experimental affinities ΔG_{exp}. (B) Histogram of the errors for pairwise relative affinities $|\Delta\Delta G_{\mathrm{pred}} - \Delta\Delta G_{\mathrm{exp}}|$. The solid line shows a Gaussian curve with a standard deviation of 1.61 kcal/mol. (C) Examples of perturbations displaying errors > 1.5 kcal/mol.

$0.64^{0.73}_{0.54}$, $\mathrm{RMSE_{pw}} = 1.69^{1.75}_{1.63}$ kcal/mol) and lower accuracy for SYK ($\tau = 0.30^{0.46}_{0.13}$, $\mathrm{RMSE_{pw}} = 2.15^{2.22}_{2.08}$ kcal/mol). Introducing features like GCMC water sampling in FEP+[81] has therefore not resolved all the challenges seen on this benchmark, and further work might be necessary.

We discuss the c-Met and SYK FEP+ results as examples in more detail. The accuracy on the c-Met data set was moderate ($\mathrm{RMSE_{pw}} = 1.43^{1.51}_{1.34}$ kcal/mol), but the predicted ΔG values show good correlation and ranking when compared to the experimental affinities (Figure 6(A)). Errors in $\Delta\Delta G$ appear to follow a Gaussian distribution (Figure 6(B)). The data set contains 12 neutral compounds and 12 positively charged compounds that carry a solvent-exposed basic group. The FEP map contained six perturbations that involved a change in net charge. For five of these transformations, the predicted $\Delta\Delta G_{\mathrm{pred}}$ was within 1.1 kcal/mol of the experimental value (examples are shown in Figure 6(C)). The remaining charge-changing transformation involved molecule CHEMBL3402762 that had a measured affinity $\mathrm{IC_{50}} < 1$ nM (top of the assay, this was set to 1 nM in order to include the compound in the comparison). This compound was predicted to be more potent than CHEMBL3402761 which has a reported $\mathrm{IC_{50}} = 1$ nM. Overall, the perturbations in the c-Met data set demonstrate that transformations involving changes in net charge can be handled reliably by the FEP+ method. Still, the benchmark cases also illustrate the challenges that are involved in predicting charge changes. In the TNKS2 set, the relative affinities within ligands of the same net charge were accurately predicted (e.g., $\mathrm{RMSE_{pw}} = 1.11^{1.18}_{1.03}$ kcal/mol for all 21 neutral ligands). The relative affinities between ligands of different net charges, however, displayed larger errors resulting in $\mathrm{RMSE_{pw}} = 2.20^{2.32}_{2.08}$ kcal/mol for the entire data set comprised of 27 ligands.

For the c-Met case, FEP+ predictions reproduced the SAR for changing from a carbamate unit to various aromatic heterocycles (Figure 6(D)). It correctly ranked pyrimidine as more potent than two thiazole variants, imidazole, oxadiazole, and pyridazine. It did, however, not reproduce the increase in potency when going from a pyridine to a pyrimidine (these two compounds that have binding affinity of IC_{50} = 200 nM and IC_{50} = 40 nM, respectively, are predicted as 49 nM and 60 nM, respectively). This change is still within the typical error limit of the method (1 kcal/mol). We initially hypothesized that the presence of the protonated species for the pyridine compound that would negatively affect the hinge binding motif and hence a difference in the protonation states of pyridine and pyrimidine compounds protonation might explain the difference in potency, but pK_a calculations with Jaguar[82] yielded a pK_a of 4.5 for the pyridine compound which makes the presence of the charged species highly unlikely (pK_a for pyrimidine compound was 1.9). Another possible explanation for the observed deviation of the predicted from the experimental value for the pyridine compound could be a lack of sampling of the rotamers of the heterocycles in the binding site. Due to its symmetry the pyrimidine compound can bind in both rotameric states contacting the hinge region of the protein via the N atom. In contrast, the pyridine can only bind in one conformation to establish this interaction. Hence, there is an entropically favorable contribution to the binding of the pyrimidine which is not taken into account if only one conformation is sampled during the simulation. Indeed, the trajectories of pyrimidine and pyridine compounds showed that the rotation was hindered in the complex simulation, whereas in the solvent both states were sampled (examples are shown in Figure S6).

For the SYK benchmark case, we found low accuracy ($RMSE_{pw} = 1.61^{1.67}_{1.54}$ kcal/mol, see Figure 7 and Table 1). This resulted in poor correlation of predicted to experimental affinities (Kendall $\tau = 0.29^{0.46}_{0.11}$). Interestingly, we saw in all simulations larger fluctuations in ligand RMSD for at least one of the ligands (>3 Å), although the compounds remained in the binding site and interactions with the hinge region were maintained. Still, this mobility might indicate difficulties in sampling.

A group of outliers was related to compound CHEMBL3265015. We observed that four out of six perturbations involving this compound displayed absolute errors >1 kcal/mol (one example is shown in Figure 7(C)). Compound CHEMBL3265015 carries a methyl group on the phenyl ring in the ortho position which might affect the rotation of the ring. Similar as in the c-Met case, sampling along this torsion angle in the complex was incomplete (data not shown).

We also noticed a large outlier when transforming from compound CHEMBL3265009 to CHEMBL3265003 ($|\Delta\Delta G_{pred} - \Delta\Delta G_{exp}| > 2$ kcal/mol, Figure 7(C)). Here, the molecule was grown by two aromatic rings extending into the solvent. Similarly, the predicted relative affinity from CHEMBL3264999 to CHEMBL3265003 was overestimated by more than 2 kcal/mol from experiment. The same results were observed for the calculations with 20 ns sampling per λ-window. A perturbation from CHEMBL3265003 to a molecule of similar size CHEMBL3265004 was, however, predicted accurately (absolute error = 0.66 kcal/mol). We observed a similar overestimation in PFKFB3 for compound **37** (see Figure S4(B)) and also for perturbations in several of our in-house projects. This overestimation of groups growing and displacing solvent at the entrance of the binding pocket might result from a water model that overestimates water mobility or underestimates water−protein interactions. Recently, several research groups found that intrinsically disordered proteins were captured better by force fields that increased the strength of the water−protein interaction.[83−85] A systematic investigation of different water models and water−protein interaction parameters could be an interesting subject of future research.

Comparing FEP+ results on the benchmark with results for Glide docking, Glide rescoring, and Prime MM-GBSA calculations, we found that FEP+ showed better or equally high correlation to experimental data for seven of the eight cases (Table 1, Table S5, and Figure 8). On this benchmark,

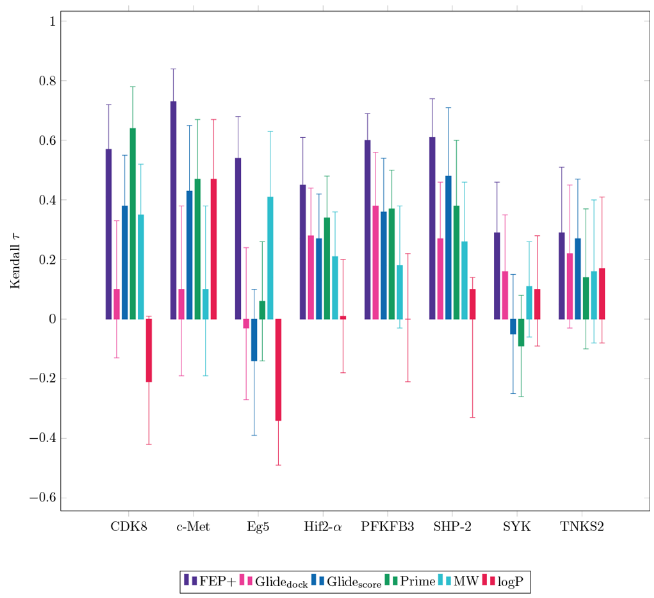

Figure 8. Comparison of Kendall τ for FEP+, Glide docking, Glide rescoring, Prime, MW, and log P ranking on the eight benchmark cases. Confidence intervals were estimated via bootstrap sampling.

FEP+ clearly outperformed the three simpler methods (Cohen's d for Kendall τ was 1.64, 1.41, and 1.41 when compared with Glide docking, Glide rescoring, and Prime, respectively, indicating a very large effect size[69,70]). In addition, it outperformed ranking by two molecular descriptors that could be considered "null models" (Figure 8 and Table S6; Cohen's d for Kendall τ was 2.38 and 2.00 for comparison with ranking by molecular weight and calculated log P, respectively, indicating a huge and very large effect size[69,70]). This result is in qualitative agreement with the data obtained from our in-house projects.

It is important to point out that the benchmark results presented in this study set were obtained in an industry context without extensive optimization following our standard protocol. It may very well be possible to obtain even higher accuracy results with more thorough model optimization. Indeed, for the Eg5 data set, we found that the protein structure−specifically one loop in the vicinity of the ligand−had a large effect on the accuracy of the prediction. We could clearly link this structural change to a set of outliers involving a phenol group that was in contact with this loop (see Figure S7). However, in the context of drug discovery projects, there is usually only limited time for validation and model optimization (typically 2 weeks). In our opinion, the performance reported here gives a realistic view to what accuracy can be expected when using FEP+ in an industry setting. We hope this set will be useful to drive further method development in the field. The input structures for the eight benchmark cases and FEP+ results for 5 and 20 ns are available on www.github.com/MCompChem/fep-benchmark (tag v1.0).

Impact of FEP+ on Projects and Operational Challenges. Despite these overall encouraging prospective FEP+ results and the clear advantage compared to simpler SBDD methods, throughout the initiative it became clear to us that the usefulness of the prediction and the required level of accuracy to achieve meaningful ranking heavily depend on the chemical series and the observed dynamic range/affinity distribution in the optimization phase. For example, for target 3/series 1 and target 3/series 3, we found good ranking with FEP+, although the reported RMSE was larger than 1.5 kcal/mol (Table S1 and Figure 3). This was helpful in prioritizing compounds for synthesis and was regarded as valuable by the project chemists. On the other hand, in target 1/series 1−3, we did not obtain any predictive ranking despite having similar accuracy in terms of RMSE. In our experience, lower accuracy can be tolerated in projects that display a larger variation in the underlying potency distribution of the chemical series. Once optimization has hit an affinity "canyon" in chemical space where small variations do not have a large effect on potency, applying free energy calculations has only limited value for the project (as was the case in target 1/series 1−3 and target 5/series 1). Interestingly, we found that such a situation was often present in (late stage) lead optimization projects. At the start of the initiative, we assumed that lead optimization should be the primary use case for FEP, since the small scale chemical transformations that are typically carried out at this stage are best suited for the method. Yet in contrast to our initial expectation, we noticed that we had better impact on hit-to-lead optimization or on fragment optimization when chemical space was more broadly explored and potency was still a major optimization parameter. Furthermore, at this point, chemistry resources were usually more limited, and chemists still faced synthetic challenges that limited the number of molecules that could be synthesized and tested experimentally. In such a situation, prioritizing ideas to focus on the most promising ones was perceived as very valuable, and performing these computationally expensive calculations was not regarded as a bottleneck. Additionally, guiding design teams toward compounds that should be made generates a higher impact than advising them which compounds should *not* be made. Assessing the impact on the optimization retrospectively for the latter is not easily done. In terms of domain of applicability, one has to keep in mind that predictivity can be limited in early hit optimization phase, since the chemical transformations encountered in these early stages are typically more challenging for the method.

Strong communication was also crucial for successfully implementing free energy calculations in projects. We experienced how important it is to have a good understanding of the capabilities and limitations of the method in the project team. This helped in selecting compounds for prioritization with FEP+ that were within the domain of applicability. Initially, we faced challenges in using FEP+ in projects because ideas submitted for calculations were outside the scope of the method. In later stages of the initiative, we focused on ranking custom-built libraries that were by design more suitable for the calculations. A disadvantage of this approach was that, in some cases, the results of such a library scan were taken up by the project team and used in the design of new molecules; however, this information was combined with other ideas. In this way, it was hard to assess the impact of the method since the exact molecules as predicted by FEP were not synthesized. To avoid this issue, we aimed to predict such molecules later with FEP+ in order to be able to assess the quality of the predictions with our automatic workflow (see above). In general, we found that the acceptance of FEP predictions by chemists was higher when the results could be interpreted or rationalized; e.g., by analyzing interactions or ligand flexibility. We therefore recommend to deliver FEP predictions accompanied by such an analysis, especially focusing on those compounds that were predicted to be the best and the worst binders.

In summary, to make best use of free energy calculations in projects, prediction accuracy, domain of applicability, key optimization challenges, synthetic accessibility of the chemical series of interest, and timing in project have to be carefully balanced. We found that screening large custom-built libraries—ideally designed with nonpotency optimization parameters in mind—to be an effective way of providing added value. For these libraries, we screen at least 50−100 ideas. We aim to screen 5−10 times more ideas than the maximum number of compounds that can be selected for synthesis (this is in line with a recent publication[86]).

■ CONCLUSION

Free energy calculations are more and more frequently used in the pharmaceutical industry and have become a powerful addition in the computational chemist's toolbox. Here, we reported data from using FEP+ prospectively in a large number of in-house drug discovery projects. We obtained good accuracy for a large variety of targets and chemical series. Yet, the number of targets for which we were able to obtain high accuracy predictions (RMSE < 1 kcal/mol) was limited. These results were in line with those obtained on a new, public benchmark set. Prepared input structures for this benchmark data set are available for the community and may drive further method development. In real world drug design projects, structural information and availability of ligand series with large affinity spread are not always given. In addition to the accuracy of the prediction, we identified multiple important operational factors that affect the impact of the method in projects. In the near future, we envision FEP+ as an expert tool to support FEP-enabled projects with large-scale library calculations.

■ EXPERIMENTAL SECTION

Protein Structure Preparation. Protein structures were downloaded from the PDB (www.rcsb.org) or from our in-house database and imported into Maestro.[87] If multiple chains were present and there was no indication that the multimer was the biologically relevant form, structures were split, and each chain was processed separately. Prime Homology modeling[88,89] was used to remove mutations introduced in the crystallization constructs (if any) and build missing side chains and missing loops. If loops had been added, these were subsequently refined with the Prime Loop Refinement protocol. Sequence alignment was performed with ClustalW and adapted manually if necessary. The homology model was built using the knowledge-based approach and taking the inhibitor into account. For loop refinement, recommended settings depending on the length of the loop were used. If larger domains (more than 20 residues) were missing, these parts of the structure were not added. If multiple structures for a given target were available, loop structures were also modeled based on alternative structures. The structures were then processed with the Protein Preparation Wizard in Maestro.[87,90] Hydrogens were added, all crystal waters were retained, and the termini were capped. Cocrystal ligand protonation states were evaluated with Epik.[91] Protein protonation states were determined with PropKa,[92,93] and hydrogen bonds were optimized. The structures were finally minimized with an RMSD cutoff of 0.3 Å (default settings).

Ligand Structure Preparation. Corina was used to generate 3D coordinates based on the 2D representation. Structures were then processed with LigPrep enumerating possible stereoisomers and assigning protonation states with Epik.[91,94] For positioning the ligands into the binding site, we either used the Flexible Ligand Alignment tool or Glide core-constrained docking[95] based on a reference structure (in most cases, the X-ray ligand). The core was defined by maximum common substructure or by a custom SMARTS pattern for molecules involving core changes. We ran Glide core-constrained docking using standard precision settings. Targets 1, 2, and 3 were prepared mainly with Flexible Ligand Alignment. For all other targets, we initially used both methods and switched almost completely to Glide core-constrained docking throughout the course of the evaluation. The poses were visually inspected, and in the case of multiple possible binding modes (e.g., for asymmetrically substituted rings), multiple possible tautomers, or charge states, binding pose FEP was used to evaluate all poses and states. The pose/state with the lowest free energy was taken as the final input structure. Structures were analyzed with the Force Field Builder to detect missing torsion parameters. If necessary, new custom parameters were generated by the Force Field Builder protocol.

Prospective Free Energy Calculations in in-House Projects. Prospective free energy calculations were performed using the Schrödinger FEP+ method[5] with Schrödinger suite versions 2016-4 to 2018-3. Up to version 2018-1, the OPLS3 force field[15] was used with custom parameters generated by the Force Field Builder. For calculations performed with version 2018-2 and higher, the OPLS3e force field[17] with custom parameters was used. Prospective calculations were run with optimal settings as determined by the validation study. In most cases, these corresponded to the default settings; in some cases, the sampling was extended beyond 5 ns per λ-window. Calculations were analyzed using the FEP+ GUI.[87] If convergence issues were found, simulations were extended if time and computing resources allowed. In all cases, unconverged edges and edges with high hysteresis values were removed from the final map. Finally, all predicted ΔG values > −5.81 kcal/mol were set to −5.81 kcal/mol to allow better comparison to experiment (typical value for bottom of the assay).

In-House in Vitro Assays. For all targets except target 8 and target 2, biochemical enzyme activity assays were used to determine the binding affinity of the molecules. For target 8, a mixture of functional and ITC measurements was used for comparison with FEP+ results (molecules that showed no activity in the functional assay were not profiled with ITC). For target 2, affinities were determined using a cellular phenotypic assay.

Free Energy Calculations on Benchmark Data Set. The full data set is summarized in Table 1. Protein cocrystal structures were downloaded from the Protein Data Bank (www.rcsb.org). Ligand structures and affinities (IC_{50}) were extracted manually from papers and patents[71−80] or extracted via ChEMBL.[96,97] Protein structures and ligand structures were prepared for free energy calculations as described above. In contrast to the in-house data sets, ligand structures were aligned by Glide core-constrained docking[95] using the X-ray ligand as reference (standard precision settings). For the benchmark, free energy calculations with FEP+ were run using Schrödinger suite version 2018-3 with the OPLS3e force field.[17] Prepared structures were loaded into the FEP+ panel, and affinity data were added. Maps were generated with default settings (optimal topology). No further modifications were made to the map. FEP+ jobs were run for 5 and 20 ns sampling time per λ-window. The output was analyzed with the FEP+ panel in Maestro. In contrast to the prospective calculations, we did not modify the final map—e.g., remove edges with high hysteresis—to allow for better reproducibility of the results.

Comparison to Other Methods. For comparison, we docked ligands into their respective protein structure using the Glide standard precision ligand docking workflow[95] in Schrödinger suite 2018-3 with default settings. As receptor, the protein structure used for the free energy calculations was used (all water molecules were deleted). The ligands were ranked according to Glide gscore and this score was compared to experimental affinities. We refer to this as the "$\text{Glide}_{\text{dock}}$" method. In addition, we simply scored the FEP+ input poses with Glide gscore. For this, we used Glide standard precision workflow with Ligand sampling "None (refine only)". All other settings were kept at default values. The FEP+ input ligand pose was minimized in the field of the receptor, and the Glide gscore score was then evaluated. We refer to this as the "$\text{Glide}_{\text{score}}$" method. MM-GBSA calculations were performed with the Prime MM-GBSA module in Schrödinger suite 2018-3 using default settings. The protein structure without water molecules was used as receptor, and the ligand poses were taken from the FEP+ input. The ligands were ranked according to the "MMGBSA dG Bind" score. Molecular weight and logP were calculated for the neutral ligands using Maestro.[87]

Analysis. Convergence, hysteresis, and interaction analysis were performed using the FEP+ GUI in Maestro. Correlations statistics and errors were evaluated using Python numpy, scipy,[98] and bootstrapped libraries. 90% symmetric confidence intervals (90% CI) for all performance metrics were calculated using bootstrap by resampling all data sets with replacement, with 10000 resampling events. CIs were estimated for all performance metrics and reported as $x_{x_{\text{low}}}^{x_{\text{high}}}$ where x is the mean statistic calculated from the complete data set (e.g., RMSE), and x_{low} and x_{high} are the values of the statistic at the fifth and 95th percentiles of the value-sorted list of the bootstrap samples. Data curation and extraction of experimental data for prospective FEP calculations from our in-house database were carried out using in-house custom scripts. IC_{50} or K_d values were converted to free energies using the equations

$$\Delta G_{\text{exp}} \approx k_B T \log IC_{50}$$

$$\Delta G_{\text{exp}} = k_B T \log K_d$$

Note that this approximation results in a constant offset of ΔG values compared to applying the Cheng-Prusoff equa-

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J. Chem. Inf. Model. 2020, 60, 5457−5474

tion.[99] This has no effect on our results which are about relative free energy ranking of compounds. However, this means the results presented here would not be directly comparable to computed absolute binding free energies.

To estimate the maximally achievable performance for each data set based on the error of the experimental binding affinity assay,[53] we generated 1,000 sets of binding free energies based on a Gaussian distribution with mean $\mu = \Delta G^{i}_{\mathrm{exp}}$ and standard deviation σ = 0.4 kcal/mol for each data point as used in previous work.[5] The average and the fifth and 95th percentiles of the resulting distribution of the calculated correlation statistics were reported as the maximally achievable performance ("Maximum"). This model reflects the expected correlation between two independent experimental measurements and their uncertainty.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jcim.0c00900.

Supplementary figures and tables, tables and supplementary tables as csv files, and ligand structures from benchmark data set as Smiles (ZIP)

■ AUTHOR INFORMATION

Corresponding Authors

Christina E. M. Schindler − *Merck KGaA, 64293 Darmstadt, Germany;* orcid.org/0000-0002-8980-048X; Email: christina.schindler@merckgroup.com

Daniel Kuhn − *Merck KGaA, 64293 Darmstadt, Germany;* Email: daniel.kuhn@merckgroup.com

Authors

Hannah Baumann − *Merck KGaA, 64293 Darmstadt, Germany*

Andreas Blum − *Merck KGaA, 64293 Darmstadt, Germany*

Dietrich Böse − *Merck KGaA, 64293 Darmstadt, Germany*

Hans-Peter Buchstaller − *Merck KGaA, 64293 Darmstadt, Germany;* orcid.org/0000-0002-0577-8319

Lars Burgdorf − *Merck KGaA, 64293 Darmstadt, Germany*

Daniel Cappel − *Schrödinger GmbH, 68161 Mannheim, Germany;* orcid.org/0000-0002-8460-7867

Eugene Chekler − *EMD Serono Research and Development Institute Inc., Billerica, Massachusetts 01821, United States*

Paul Czodrowski − *Merck KGaA, 64293 Darmstadt, Germany;* orcid.org/0000-0002-7390-8795

Dieter Dorsch − *Merck KGaA, 64293 Darmstadt, Germany*

Merveille K. I. Eguida − *Merck KGaA, 64293 Darmstadt, Germany*

Bruce Follows − *EMD Serono Research and Development Institute Inc., Billerica, Massachusetts 01821, United States*

Thomas Fuchß − *Merck KGaA, 64293 Darmstadt, Germany*

Ulrich Grädler − *Merck KGaA, 64293 Darmstadt, Germany*

Jakub Gunera − *Merck KGaA, 64293 Darmstadt, Germany*

Theresa Johnson − *EMD Serono Research and Development Institute Inc., Billerica, Massachusetts 01821, United States*

Catherine Jorand Lebrun − *EMD Serono Research and Development Institute Inc., Billerica, Massachusetts 01821, United States*

Srinivasa Karra − *EMD Serono Research and Development Institute Inc., Billerica, Massachusetts 01821, United States*

Markus Klein − *Merck KGaA, 64293 Darmstadt, Germany*

Tim Knehans − *Merck KGaA, 64293 Darmstadt, Germany*

Lisa Koetzner − *Merck KGaA, 64293 Darmstadt, Germany*

Mireille Krier − *Merck KGaA, 64293 Darmstadt, Germany*

Matthias Leiendecker − *Merck KGaA, 64293 Darmstadt, Germany*

Birgitta Leuthner − *Merck KGaA, 64293 Darmstadt, Germany*

Liwei Li − *EMD Serono Research and Development Institute Inc., Billerica, Massachusetts 01821, United States*

Igor Mochalkin − *EMD Serono Research and Development Institute Inc., Billerica, Massachusetts 01821, United States*

Djordje Musil − *Merck KGaA, 64293 Darmstadt, Germany*

Constantin Neagu − *EMD Serono Research and Development Institute Inc., Billerica, Massachusetts 01821, United States*

Friedrich Rippmann − *Merck KGaA, 64293 Darmstadt, Germany;* orcid.org/0000-0002-4604-9251

Kai Schiemann − *Merck KGaA, 64293 Darmstadt, Germany*

Robert Schulz − *Merck KGaA, 64293 Darmstadt, Germany; Institute of Pharmacy, Freie Universität Berlin, 14195 Berlin, Germany*

Thomas Steinbrecher − *Schrödinger GmbH, 68161 Mannheim, Germany*

Eva-Maria Tanzer − *EMD Serono Research and Development Institute Inc., Billerica, Massachusetts 01821, United States*

Andrea Unzue Lopez − *Merck KGaA, 64293 Darmstadt, Germany*

Ariele Viacava Follis − *EMD Serono Research and Development Institute Inc., Billerica, Massachusetts 01821, United States*

Ansgar Wegener − *Merck KGaA, 64293 Darmstadt, Germany*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.jcim.0c00900

Notes

The authors declare the following competing financial interest(s): D.C., T.K., and T.S. are employees of Schroedinger GmbH.

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